INCH-POUND
MIL-DTL-24643/12E
22 August 2002
SUPERSEDING
MIL-C-24643/12D
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, 300 VOLTS, TYPE LSTTOP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First -	Copper conductor - AWG size No. 20, class K stranding, uncoated (see note 1).
Second -	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third -	Ethylene propylene rubber or cross-linked polyethylene insulation. Minimum average wall thickness 0.020 inch. Telephone identification code shall be applied by method 3.
Fourth -	Two conductors shall be cabled together to form pairs, with a lay suitable to meet test requirements specified herein.
Fifth -	The specified number of pairs (see table I) shall be cabled together in one or more layers, all in left hand direction, with the minimum amount of fillers necessary, to obtain a firm well-rounded assembly. Cabling sequence to be consecutive, starting with No. 1, from center outward. The length of lay shall be determined to permit a completed cable to meet the twisting and bending endurance tests. A single pair shall not be used as center for the cable core.

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Construction (non-watertight) (continued)

Sixth - Binder tape applied helically with overlap.

Seventh- Cross-linked polyolefin jacket. (see table I for wall thickness)

NOTE: 1. At the manufacturer's option, conductors may consist of eight strands, each having a nominal diameter of 0.010 inch, around one strand having a nominal diameter of 0.0142 inch.

TABLE I. Details.

Military part no. M24643/12	Type and size	Number of pairs	Cable jacket thickness min avg (inch)	Overall diameter maximum (inch)
-01UN	LSTTOP-3	3	0.060	0.480
-02UN	LSTTOP-5	5	0.060	0.590
-03UN	LSTTOP-10	10	0.060	0.700
-04UN	LSTTOP-15	15	0.060	0.830

EXAMINATION AND TESTS:

Basic electrical: Requirements:

Conductor resistance - ohms/1000 feet at 25°C, maximum 11.53
Voltage withstand - volts, root mean square, minimum
conductor to conductor 2000
Insulation resistance - conductor to conductor,
Megohms -1000 feet, minimum.....100
Conductor continuity ..... No failure

Group A:

Visual and dimensional..... No failure

Group B:

Cross-linked proof test (percent, maximum) Insulation 50
Jacket (When tested at 200°C).....50
Tear - pounds per inch thickness, minimum (ASTM D 470) 35

Physicals (unaged)

Insulation

Ethylene propylene rubber

Tensile strength - lb/in^2, minimum 700
Elongation - percent, minimum 250

Cross-linked polyethylene

Tensile strength - lb/in^2, minimum 1800
Elongation - percent, minimum 250

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Jacket (cable)
Tensile strength - lb/in^2, minimum 1300
Elongation - percent, minimum 160
Twisting and bending endurance - (see note 2)
At 75°C, cycles, minimum.....6000
At minus 30 ± 2°C, cycles, minimum.....2000

NOTE: 2. Roller diameter for bending endurance test shall be as follows:

Cable type and size	Roller diameter (inches, maximum)
LSTTOP-3	1.4
LSTTOP-5	1.8
LSTTOP-10	2.1
LSTTOP-15	3.0

Group C:
Physicals (aged)
Insulation
Ethylene propylene
Air oven
Tensile strength - percent of unaged, minimum 75
Elongation - percent of unaged minimum 75
Cross-linked polyethylene
Air oven
Tensile strength - percent of unaged, minimum 80
Elongation - percent of unaged minimum 80
Jacket (cable)
Air oven
Tensile strength - percent of unaged, minimum 60
Elongation - percent of unaged minimum 60
Hot oil immersion
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged minimum 50
Shrinkage ..... No failure
Permanence of printing (jacket) - cycles, minimum.....125
Heat distortion - percent of unaged, maximum 30

Group D:
Flame propagation (cable) ..... No failure

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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Size	Feet (nominal)
3, 5 and 10	2500
15	1500

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-010)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

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